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A hydride model for solid and liquid cerium ejecta particles

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Cerium ejecta particles form after a shock wave impacts a cerium plate with a rough surface or prescribed surface perturbation. When these particles are ejected into hydrogen or deuterium gas the material reacts exothermically and raises the temperature of the particle. Two models are developed to capture the heat and mass transfer that occurs during this process. One model assumes that for larger solid ejecta particles, the loss of mass is negligible while for smaller liquid ejecta particles, the loss of mass during reaction is accounted for. The loss of mass is assumed to occur from the hydride layer being extremely brittle and the fact that it will be relatively easy for the hydride material on the outside of the particle to go into tension, crack and flake off. The model is tested by performing simulations similar to the original experiments that motivated the development of these models and comparing the particle temperatures. While the model is able to capture some general features of the flowfield, inaccuracies clearly exist that must be addressed and the results underscore the need for more isolated experiments where fewer assumptions and physical processes are present simultaneously.

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INTRODUCTION

Ejecta form from Richtmyer-Meshkov (RM) [1, 2] unstable growth of surface perturbations after being impacted by a shock wave. Analytic models exist [3, 4] to describe the evolution of spikes and bubbles that originate from linear surface perturbations (2D unstable sheets) or from single dots (3D spikes). These spikes eventually fragment [5–9] into individual particles that travel at reasonably well known velocities [4, 10–14]. The amount of ejecta created is a function of the surface perturbation amplitude and whether the surface is liquid or solid after shock release [15].

The ejecta size/mass and velocity distributions have been studied in both gases and vacuum [16–28]. It is known [29] that droplets with sufficiently small Ohnesourge numbers and Weber numbers below a critical value of 11 are relatively stable and do not breakup much further.

In 2017 ejecta experiments [30, 31] ejecting both tin and cerium into D_2 gas were performed with the goal of comparing inert and reactive ejecta transport using liquid droplets. However, the performed cerium experiments produced solid ejecta particles due to errors in the EOS used to compute the required shock strength to cause the cerium to fully melt upon shock release. The exothermic reaction of the cerium in these experiments raises the temperature of the particles relative to their initial temperature as well as relative to the gas temperature by up to 400-500 K. In these experiments [30, 32] the tin particles are liquid with an observed mean diameter of 2 μm while the cerium particles are solid with a mean diameter of 13μm. In the solid cerium experiments ejecting into D_2 gas the particle radiance reached a maximum 5μs after shock release from the target surface, after which the radiance temperature declined. The velocity profiles obtained via photon Doppler velocimetry (PDV) decreased monotonically in a manner similar to the tin experiments.

Then in 2019 [31], additional experiments were performed with cerium into D_2 gas where liquid ejecta particles were created with a mean size of 2 μm. In these experiments, the temperature increases substantially faster, but then plateaued thereafter. Additionally, the velocity in the PDV traces appeared erratic, almost as if the velocity was not decreasing monotonically. The strange velocity behavior is attributed to the impact of the reactions on much smaller particles. It is hypothesized that the key difference between the solid and liquid ejecta experiments lies in the size of the particles and the propensity for the smaller liquid particles to shed small pieces of the hydride layer into the gas as sub-micron sized particles.

A wide range of particle combustion models exist and are summarized in [33–35] among others.

High ash content coal combustion models assume that a layer of ash grows around the outer surface of a reacting coal particle as the char-ash interface moves closer to the particle center. The reaction proceeds as a non-volatile heterogeneous surface reaction. Schwarzkopf *et al.* [36] developed a model for cerium hydriding in deuterium gas that is very similar to the 1-D high ash content model described in [35] where the chemical reaction is controlled by the rate of diffusion through the outer hydride layer. Regele *et al.* [37] later expanded on this model.

It is suspected that a combination of the lower density of the hydride layer relative to the metal core combined with the reaction causes the particle to expand and puts the hydride layer in tension. The induced tension would then cause the surface of the hydride layer to crack and potentially flake off small pieces. While the constantly growing shell model developed previously [37] may be reasonably accurate for the larger solid ejecta particles, where the mass of the flaked shell may be a small fraction of the total ejecta mass, it is most likely not accurate in the smaller liquid ejecta experiments and may be responsible for the strange PDV traces.

In this work, the previous particle hydride models are expanded on to account for the effects of hydride layer flaking as well as reaching the melt temperature of the hydride layer. Simulations that mimic the original experiments are performed and comparisons are made.

I. MATHEMATICAL MODEL

The evolution of cerium ejecta particles is a function of a number of different physical processes, including chemical reactions, heat and mass transfer. In order to model the hydriding process of cerium particles in a deuterium gas, a number of assumptions must be made to simplify the physics involved for the particle:

- the ejecta particles are hydrodynamically stable and spherical
- deuterium must diffuse through a layer of cerium dihydride before reaching the unreacted cerium
- deuterium diffuses through the dihydride layer in a quasi-steady manner such that the reaction at the pure cerium surface occurs rapidly
- heat conducts much faster inside the particle relative to the surrounding gas so that the particle temperature can be assumed to be constant.

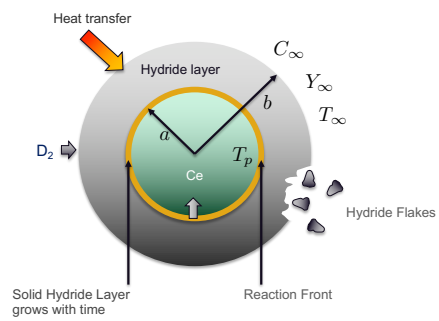


FIG. 1. Diagram of an ejecta particle with pure cerium surrounded by a layer of cerium dihydride.

These assumptions lead to the model illustrated in Fig. 1 where cerium is located in the center of the particle with radius a and a solid dihydride layer surrounds that region with radius b. Deuterium must diffuse through the dihydride layer until it reacts rapidly with the cerium. As the cerium reacts, the volume of cerium radius a retreats as the dihydride layer advances. The particle is at a constant temperature T_p and the surrounding gas has a concentration C_∞ or mass fraction Y_∞ and temperature T_∞. It is possible for the hydride surface to go into tension due to the overall particle undergoing expansion from the reaction and since cerium dihydride has a fluorite structure as a solid, it is very brittle and will likely flake off small pieces as it reacts.

A. Molecular diffusion

It is assumed that the particles are spherical, hydrodynamically stable and do not hydrodynamically breakup any further. The deuterium gas diffuses through a solid layer of cerium dihydride where the gas concentration far away from the particle is C_∞ or the mass fraction is Y_∞. The time unsteady molecular diffusion mass fraction equation describing this process is expressed as

$$\frac{\partial Y}{\partial t} = \nabla \cdot \mathscr{D} \nabla Y \tag{1}$$

where Y is the mass fraction of deuterium anywhere inside the particle and $\mathscr{D}$ is the diffusion coefficient of deuterium through the outer, solid layer of cerium dihydride which is being formed as the product of reaction. If the particle is hydrodynamically stable, it can be assumed that the particles are at least close to spherical in nature and the Laplacian is limited to the radial direction. Since it can be accurately assumed that the particle temperature is uniform inside the particle, radial variations in the diffusion coefficient will be negligible, allowing $\mathscr{D}$ to be removed from the

radial partial derivative, giving

$$\frac{\partial Y}{\partial t} = \mathscr{D}\left(\frac{\partial^2 Y}{\partial r^2} + \frac{2}{r}\frac{\partial Y}{\partial r}\right). \tag{2}$$

Finally, it can be reasonably assumed that the reaction at the cerium surface occurs fast enough that the reaction is diffusion controlled rather than controlled by kinetics and the concentration of deuterium at the cerium/cerium dihydride interface is negligible ($Y_s = 0$). This is a common assumption in particle combustion models [33–35] and will be employed here. The resulting differential equation becomes

$$\mathscr{D}\frac{d}{dr}\left(r^2\frac{dY}{dr}\right) = 0\,. \tag{3}$$

The solution for the mass fraction as a function of radius is

$$Y(r) = Y_\infty \frac{ab}{a-b}\left(\frac{1}{r} - \frac{1}{a}\right). \tag{4}$$

The mass conversion rate of deuterium reacting with the pure cerium surface is evaluated to be

$$\dot{m}_{D_2} = -\rho_{D_2}\mathscr{D}\,4\pi r^2 \frac{dY}{dr}\bigg|_{r=a} = 4\pi\mathscr{D}ab\frac{\rho_\infty Y_\infty}{b-a}\,. \tag{5}$$

Here ρ_{D_2} is the density of the gas inside the hydride layer, ρ_∞ is the ambient gas density, Y_∞ is the ambient gas mass fraction, b is the particle outer radius and a is the radius of the cerium/cerium dihydride interface. It's clear that the mass conversion rate $\dot{m}_{D_2} \to \infty$ when $b = a$. When $b = a$ the hydride layer has zero thickness, which suggests that the rate limiting step at that point in time would be the chemical kinetics on the initially unreacted cerium surface. Since the current model does not account for these initial kinetics dominated reactions, the mass conversion rate employed here will overestimate the conversion rate by a small amount.

B. Reaction model

Cerium reacts with deuterium to form CeD_2 after which it can continue to produce CeD_3. The second reaction is slower than the first and doesn't always run to completion so that the final reaction product becomes CeD_{2+x}, where $x \in [0,1]$ [38]. The details of determining the final product composition is beyond the scope of the present work, thus only the first reaction is considered at this time. The reaction can then be written

$$\mathrm{Ce} + \mathrm{D_2} \to \mathrm{CeD_2}\,, \tag{6}$$

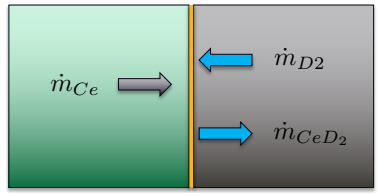


FIG. 2. Diagram illustrating the movement of cerium, deuterium and cerium dihydride relative to the reaction interface highlighted in yellow. Pure cerium is on the left in green and deuterium diffuses through the solid cerium dihydride before reaching the reaction interface.

where one deuterium molecule reacts with each cerium atom. In this instance, the mass stoichiometric coefficient for this reaction is written as a ratio of molecular weights

$$i = \frac{W_{Ce}}{W_{D_2}}, \tag{7}$$

which can also be related to the ratio of mass conversion rates $i = \dot{m}_{Ce}/\dot{m}_{D_2}$. The mass stoichiometric coefficient can be used to rewrite Eq. 6 in terms of mass so that

$$(i\,\mathrm{kg})\,\mathrm{Ce} + (1\,\mathrm{kg})\,\mathrm{D_2} \rightarrow (i+1\,\mathrm{kg})\,\mathrm{CeD_2}\,. \tag{8}$$

Now the mass consumption rate of cerium can be directly related to the mass consumption rate of deuterium so that $\dot{m}_{Ce} = i\dot{m}_{D_2}$.

Figure 2 illustrates the flow of the reactant and product species relative to the reaction front highlighted in yellow. Deuterium diffuses through the solid cerium dihydride layer and reacts instantly upon reaching the cerium surface. Mass conservation relative to the reaction front is written

$$\dot{m}_{CeD_2} = \dot{m}_{Ce} + \dot{m}_{D_2}\,. \tag{9}$$

The mass rate of change of cerium inside the particle is written as

$$\dot{m}_{Ce} = \rho_{Ce} 4\pi a^2 \frac{da}{dt}\,. \tag{10}$$

The mass stoichiometric coefficient (Eq. 7) is used to equate the cerium (Eq. 10) and deuterium (Eq. 5) mass conversion rates to obtain an ordinary differential equation for the reaction radius

$$\frac{da}{dt} = -i\frac{\rho_\infty}{\rho_{Ce}}\frac{Y_\infty \mathscr{D}}{b-a}\frac{b}{a}\,. \tag{11}$$

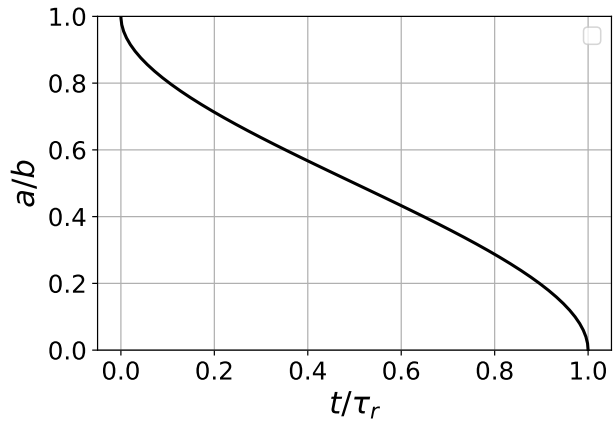


FIG. 3. Solution to Eq. 11 illustrating the change in reaction radius from $t = 0$ to $t = \tau_r$.

Figure 3 plots the solution to the ODE for constant particle radius b with time normalized by the characteristic reaction time, which will be defined in Sec. II A 1. The inner radius changes rapidly when a is initially equal to b and when $a \to 0$. One-dimensional Cartesian models [35] do not have the b/a term on the right and thereby do not capture the reaction speedup that occurs for small values of a. Otherwise, planar 1-D models are nearly identical to this model for most of the particle reaction period.

Note that both the reaction radius a and the outer particle radius b are free to change with time. If b is assumed to be constant, this equation can be integrated into a parametric expression for $t = t(a)$. Otherwise it can be solved numerically.

In general, the outer particle radius b changes with time. Conservation of mass for the particle provides the particle outer diameter

$$b^3 = \frac{3m_p}{4\pi\rho_{CeD_2}} + a^3\left(\frac{\rho_{CeD_2} - \rho_{Ce}}{\rho_{CeD_2}}\right), \tag{12}$$

where the rate of change of the particle mass is $\dot{m}_p = \dot{m}_{D_2}$. Both m_p and a are functions of time. The particle mass m_p increases with time as the particle absorbs the surrounding gas. The second term on the right side of Eq. 12 is proportional to the density difference and a^3. Since $\rho_{CeD_2} < \rho_{Ce}$, this term is negative and the particle diameter increases as a decreases. In the limit that the two densities are equal there will be no contribution from this term.

C. Heat transfer model

It is common in many particle combustion models [33, 35] to use the lumped capacitance model, which assumes that thermal conduction inside the particle is much faster than convection into the gas so that the particle temperature can be assumed to be uniform. Radiation is assumed to be small relative to convection since the difference in temperature between the two phases is assumed to be small. Convective heat transfer at the particle surface is expressed [39]

$$\dot{Q}_c = Nu\pi k_c 2b(T_\infty - T_p)\,, \tag{13}$$

where $\dot{Q}_c$ is the heat transfer rate, $Nu = \frac{hb}{k_c}$ is the Nusselt number, h is the convective heat transfer coefficient, k_c is the thermal conductivity of the continuous phase (gas), b is the outer radius, T_∞ is the ambient gas temperature and T_p is the particle temperature. The Ranz-Marshall correlation [40] is a common model for forced convection over spherical particles [39] for Reynolds numbers up to 5 x 10^4 and is written as

$$Nu = 2 + 0.6Re_r^{1/2}Pr^{1/3}\,, \tag{14}$$

where Re_r is the particle Reynolds number using the velocity of the particle relative to the gas and Pr is the Prandtl number.

D. Energy Conservation

Energy conservation for a reacting particle using the growing shell model with heat transfer can be expressed

$$\dot{E}_p = \dot{Q}_r + \dot{Q}_c\,, \tag{15}$$

where $\dot{E}_p$ is the rate of change of the particle internal energy, $\dot{Q}_r$ is the heat release from reaction, and $\dot{Q}_c$ is the heat transferred to the particle from the gas given in Eq. 13.

The heat release from reaction can be expressed

$$\dot{Q}_r = \dot{m}_{Ce}h_r\,, \tag{16}$$

where h_r is the heat of reaction or the amount of energy per unit mass of cerium released during a chemical reaction. The heat of reaction is expressed as the difference of product and reactant

enthalpies, i.e.

$$h_r = h_{prod} - h_{reac}\,, \tag{17}$$

where according to Eq. 6, h_{prod} is cerium dihydride's enthalpy of formation h_f^0 and $h_{reac} = 0$ for both pure cerium and deuterium at standard temperature and pressure.

The energy rate of change inside the particle can be written as

$$\dot{E}_p = m_p c^m \frac{dT_p}{dt}\,, \tag{18}$$

where c^m is the particle specific heat, dT_p/dt is the particle temperature rate of change, and the particle mass is the sum of all masses inside the particle $m_p = m_{Ce} + m_{CeD_2}$. Again, the contribution from nonzero $\dot{m}_p$ is neglected because the mass of deuterium required for reaction is so small relative to cerium. Summation of the different terms in Eq. 15 yields

$$\frac{dT_p}{dt} = \frac{Nu}{2}\frac{1}{\tau_T}(T_\infty - T_p) + \frac{\dot{m}_{Ce} h_r}{m_p c^m}\,, \tag{19}$$

where the thermal response time [39] is defined as

$$\tau_T = \frac{\rho_p c^m b^2}{3k_c}\,, \tag{20}$$

where ρ_p is the mean particle density m_p/V_p, c^m is the mean particle specific heat and k_c is the thermal conductivity of the continuous phase (gas).

E. Shedding Model

As discussed by Buttler [31], the reactive, liquid ejecta exhibit some manner of break up phenomena during their transport in recorded velocimetry data. To model this, modifications are needed to the growing shell model described in the previous sections. The Weber number, defined as $\frac{\rho_f |u_f - u_p|^2 d_p}{\sigma}$ with ρ_f and u_f being the fluid density and velocity, u_p being the particle velocity and σ being the surface tension, is the dimensionaless parameter representing the ratio of hydrodynamic and surface tension forces on a liquid droplet and is used in the analysis of hydrodynamic droplet breakup. Preliminary calculations of the Weber number of the ejecta droplets seen in the experiments computed Weber numbers on the range of 10^{-3} to 10. These values are generally too small to indicate that hydrodynamic break up or atomization is the primary mechanism for ejecta break up, especially on the time scales seen in these experiments. As described in the opening

to Section I, cerium dihydride has a flourite structure as a solid and is very brittle and prone to flaking off small pieces as it reacts and transports in a fluid. These facts combined motivated the development of a model to account for hydride shell shedding as the primary means of hydriding ejecta break up.

First, an assumption is made that any shell flaking (i.e. shedding) effects on the overall ejecta dynamics or thermodynamics will only be appreciable if the total mass lost due to shedding in an ejecta cloud is a non-negligible portion of the overall cloud mass. In theory, the mechanical behavior of cerium dihydride in a cross-flow should be insensitive to the size of the cerium ejecta from which it forms. What changes is the mass fraction of shed hydride in the resulting flow. As a quantitative example, suppose that two cerium particles of diameters 10 and 1 μm are taken which have reacted to the point of having hydride shells of thickness 50 nm. In the extreme case of the full shells both flaking off the ejecta particles, the larger particle would shed off a mass of roughly $4.3x10^{-14}$g while the smaller particle would shed a mass of roughly $4.5x10^{-16}$g which are 1.2 % and 12.9 % of the particle masses respectively. Thus, assuming that the spread of particle sizes inside of an ejecta cloud is not too wide, the mass of the shed shell is only a substantial amount of the ejecta cloud mass as the initial size of the ejecta become smaller. Thus, this model should only be utilized if the initial ejecta sizes are expected to be rather small, otherwise the shed mass will be nearly, if not completely, negligible and the growing shell model is a valid approximation for the resulting flow physics. For initial purposes, we take this threshold to mean that the model should be utilized if the ejecta diameters are expected to be $O(1\mu m)$ since particles of larger diameters ($O(10\mu m)$ or above) are expected to contain less than 1% by mass of shed hydride in their transporting clouds. This threshold is met by the liquid cerium ejecta experiments performed by Buttler but is not met by the earlier, solid cerium ejecta experiments.

The shedding model has two major components. The first is a modification to the reaction model described in Section I B. The preliminary version of the shedding model works by assuming that the hydriding reaction proceeds normally up until the thickness of the hydride shell, denoted by δ, reaches a given value. At this value, the shell is assumed to have reached a size where the outermost portion of the shell is in tension and can physically begin to crack and flake off due to stresses in the shell. This flaking process in reality will be spherically asymmetric and will also lead those portions of the shell to be thinner until the shell is allowed to grow back. At this time, modeling all of these synchronous effects at a subgrid scale is not feasible, so the model approaches the dynamics in a time and azimuthally averaged manner. This is done in the model by

assuming δ is an effective shell thickness that is known a priori, which is constant as the reaction progresses. The exact value which δ should take is unknown, but it is assumed to be likely in the sub-micron range. The shell mass lost due to utilizing this assumption is taken to be the shell which has flaked off into the carrier fluid. At this stage, mathematically, this imposes the constraint that $b = a + \delta$ where δ is a constant. This constraint can be used to rewrite Eq. 11 as

$$\frac{da}{dt} = -iY_\infty \mathscr{D} \frac{\rho_\infty}{\rho_{Ce}} \frac{a+\delta}{a\delta}. \tag{21}$$

Since the assumption for this differential equation is that δ is constant and not b, this equation has no closed-form, parametric analytic solution such as the one which exists for Eq. 11. It should be noted that an implicit analytic solution for the reaction radius does exist for Eq. 21, however that solution requires the use of a non-linear root-finder to obtain the solution. Thus, for computational efficiency, this equation is integrated numerically when in use. Any shell mass computed to be lost while solving this equation is subsequently given back to the carrier fluid as a species operating as a dusty gas in an additional coupling mechanism between the ejecta and the fluid.

The second modification is the addition of a phase change model for the cerium hydride shell on top of the shedding model. This model is added due to evidence from radiance temperature data [31] which suggests that the liquid reactive ejecta eventually reach a temperature plateau indicative of a melting phase change near the theorized melt point of CeD_2. The phase change alters the energy conservation morphology of the previous section. In general, this model works for both melting and freezing of the shell, with freezing just being the negation of the melting process but, for simplicity and brevity, the below discussion will just focus on a melting process. The melting process is of special importance to the model as it is assumed that a fully melted hydride shell will also lead to the halting of any shell shedding processes due to the unlikeliness of hydrodynamic breakup occurring in the single-shock driven ejecta system described at the start of the section.

The energy input required to melt a spherical hydride shell surrounding a cerium particle at the onset of melting is given by

$$H_{fus} = h^o_{fus} m_{shell} = \frac{4}{3}\pi h^o_{fus} \rho_{CeD_2}(b^3 - a^3) \tag{22}$$

where h^o_{fus} is the latent heat of fusion and all terms are evaluated at the time the particle temperature reaches the melt point of the shell ($T_p = T_m$). From this, the rate of change of particle internal

energy due to energy consumption used to drive the phase change can be written as

$$\dot{Q}_{phase} = \frac{1}{m_p}\frac{dH_{fus}}{dt} = \frac{1}{m_p}\left[4\pi h^o_{fus}\rho_{CeD_2}(b^2-a^2)\frac{da}{dt}\right]. \tag{23}$$

Note that, during a phase change in a material, the temperature of the material is held fixed while the phase change process occurs. Due to Eq. 18, this also means that the specific internal energy is also fixed during this time. Thus, during phase change, we have that

$$\dot{E}_p = \dot{Q}_r + \dot{Q}_c + \dot{Q}_{phase} = 0 \tag{24}$$

and that the energy used for the phase change balances the energy obtained via convection and reaction. However, outside of a phase change occuring, $\dot{Q}_{phase} = 0$, and Eq. 15 holds as before. So, the role of the phase change model is to modify Eq. 15 to become

$$\dot{E}_p = \dot{Q}_r + \dot{Q}_c + \delta_D(T_p - T_m)H(H_{fus})\dot{Q}_{phase} \tag{25}$$

where δ_D is the Dirac delta function and H is the Heaviside function. The $H(H_{fus})$ term is included to allow the energy conservation to revert back to Eq. 15 once the energy requirement for the phase change is achieved. Thus, the shedding model's interpretation of the particle temperature equation (Eq. 19 previously) is

$$\frac{dT_p}{dt} = \frac{Nu}{2}\frac{1}{\tau_T}(T_\infty - T_p) + \frac{\dot{m}_{Ce}h_r}{m_p c^m} - \delta_D(T_p - T_m)H(H_{fus})\frac{1}{m_p^2 c^m}\dot{Q}_{phase}. \tag{26}$$

The details on how the model implementation numerically solves the phase change portion of the above equation in a simulation will be described in a later section.

F. Model summary

To summarize, the particle hydride model is described with mass and energy conservation equations where mass conservation is expressed as an ordinary differential equation for the cerium particle radius (Eq. 11)

$$\frac{da}{dt} = -i\frac{\rho_\infty}{\rho_{Ce}}\frac{Y_\infty \mathscr{D}}{b-a}\frac{b}{a} \tag{27}$$

and energy conservation, outside of any phase change process in the shell, is written in terms of the particle temperature (Eq. 19)

$$\frac{dT_p}{dt} = \frac{Nu}{2}\frac{1}{\tau_T}(T_\infty - T_p) + \frac{\dot{m}_{Ce}h_r}{m_p c^m} - \delta_D(T_p - T_m)H(H_{fus})\frac{1}{m_p^2 c^m}\dot{Q}_{phase}. \tag{28}$$

The cerium mass consumption rate is found by combining Eq. 5 with Eq. 7 to obtain

$$\dot{m}_{Ce} = i4\pi ab\mathscr{D}\frac{\rho_\infty Y_\infty}{b-a}\,. \tag{29}$$

The numerical solution to Eq. 27 is dependent on the choice of the growing shell or shedding models. It is assumed that an appropriate momentum conservation equation will be solved simultaneously with these two conservation laws.

II. NUMERICAL INTEGRATION METHOD

It is more convenient to solve Eq. 28 in terms of particle internal energy instead of temperature. Eq. 28 can be written in terms of the particle's internal energy

$$\frac{de_p}{dt} = \frac{Nu}{2}\frac{c^m}{\tau_T}(T_\infty - T_p) + \frac{\dot{m}_{Ce}h_r}{m_p} - \delta_D(T_p - T_m)H(H_{fus})\frac{1}{m_p^2}\dot{Q}_{phase}\,. \tag{30}$$

Note that both Eq. 27 and 29 contain roots where real values of $a \in [0,b]$ may lead to division by zero without proper treatment.

Preliminary first order numerical integration schemes of Eq. 27 required $O(10^5)$ timesteps per particle to obtain particle temperatures that were relatively insensitive to timestep size. Upon closer examination, it was determined that Eq. 27 could be integrated analytically. An exact solution remedies both of the potential division by zero scenarios in 27, but there is still the possibility of division by zero in Eq. 29. Section II A presents how the analytical solution is found and the assumptions made to obtain it. Section II B shows how the energy equation is discretized along with the mass conversion rate (Eq. 29).

A. Reaction Radius Integration

1. Growing Shell Model

In order to obtain an analytical solution to Eq. 27, it is helpful to simplify the equation into a simple mathematical form. The first step is to define a characteristic reaction time for the growing shell model, namely

$$\tau_r = \frac{1}{6i}\frac{\rho_{Ce}}{\rho_{D_2}}\frac{b^2}{\mathscr{D}Y_\infty}\,. \tag{31}$$

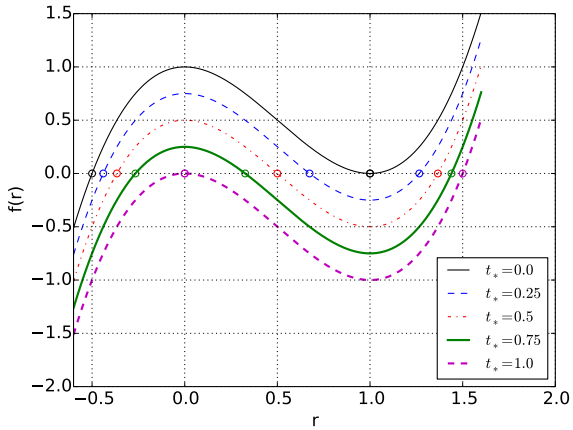


FIG. 4. Solution to Eq. 34 for several values of t_* illustrates the middle root gives real roots for $0 < t_* < 1$.

This solution is obtained by integrating Eq. 27 to obtain $t(a)$ as a cubic polynomial with b constant and then evaluating $t = \tau_r$ with $a = 0$. Note that this time is $1/3$ the time obtained assuming a planar diffusion model because the spherical coordinate system causes a $1/a$ term to appear in Eq. 27, which speeds up the reaction when the particle diameter becomes small.

The next step is to transform the reaction radius equation using the reaction time (Eq. 31) and the non-dimensional reaction radius $r = a/b$ so that Eq. 27 becomes

$$\frac{dr}{dt} = \frac{1}{6\tau_r}\frac{1}{1-r}\frac{1}{r}\,. \tag{32}$$

For the moment, it is assumed that every term in Eq. 27 is constant except for the reaction radius a so that τ_r is constant. In reality, the particle radius b will change, which will lead to changes in τ_r. Under these conditions, Eq. 32 can be integrated to give

$$\frac{t}{\tau_r} = 2r^3 - 3r^2 + 1\,, \tag{33}$$

where $r = 1$ when $t = 0$ and $r = 0$ when $t = \tau_r$. Finally, substitution of $t_* = t/\tau_r$ gives

$$0 = 2r^3 - 3r^2 + (1 - t_*) \tag{34}$$

where $t_* \in [0,1]$ and $r \in [0,1]$.

This cubic equation must be solved to obtain the reaction radius r. Figure 4 plots this function for several values of t_*. Visual inspection of the solution for the valid range of t_* shows that the middle root provides the real solution for $0 < t_* < 1$ and double roots when $t_* = 0$ and $t_* = 1$. A

trigonometric solution method gives the solution as

$$r(t_*) = \cos\left(\frac{2\pi}{3} - \theta(t_*)\right) + \frac{1}{2} \tag{35}$$

with

$$\theta(t_*) = \frac{1}{3}\arccos(2t_* - 1)\,. \tag{36}$$

This solution assumes that τ_r is constant over the entire reaction period. However, recall that the solution will be used in conjunction with a hydrodynamic fluid solver that integrates in time with a step Δt where τ_r is only assumed to be constant over Δt. The non-dimensional change in radius from timestep n to $n+1$ with $t_*^{n+1} = t_*^n + \Delta t_*$ can be expressed

$$\Delta r_n^{n+1} = r(t_*^{n+1}) - r(t_*^n) \tag{37}$$

$$= r^{n+1} - r^n \tag{38}$$

$$= \cos\left(\frac{2\pi}{3} - \theta^{n+1}\right) - \cos\left(\frac{2\pi}{3} - \theta^n\right)\,. \tag{39}$$

Transformation back into dimensional variables gives the reaction radius a at $t^n + \Delta t$ as

$$a^{n+1} = a^n + b\Delta r_n^{n+1} \tag{40}$$

with

$$\theta^n = \frac{1}{3}\arccos\left(2\frac{t^n}{\tau_r^n} - 1\right) \tag{41}$$

$$\theta^{n+1} = \frac{1}{3}\arccos\left(2\left[\frac{t^n + \Delta t}{\tau_r^n}\right] - 1\right)\,. \tag{42}$$

Equations 40, 41 and 42 provide the reaction radius at $t^{n+1} = t^n + \Delta t$ given the reaction radius a^n at $t = t^n$.

2. *Shedding Model*

The shell-shedding model handles the numerical solution to Eq. 27 in a slightly different manner. For times t such that the outer radius of the ejecta particle is less than the sum of the reaction radius and the imposed hydride shell thickness (i.e. $b(t) < a(t) + \delta$), the above analytic solution approach is still used. However, once the condition $b(t) > a(t) + \delta$ is achieved, the assumption of a constant outer radius for the particle used to obtain the analytic solution to Eq. 27 is no longer

valid. At this stage, the shedding model reverts to a first-order, implicit time-marching scheme to solve Eq. 27 as described in Section I E.

This implicit discretization, when applied to Eq. 27, results in the following form

$$\frac{a^{n+1}-a^{n}}{\Delta t} = -C\frac{a^{n+1}+\delta}{\delta a^{n+1}} \tag{43}$$

where $C = iY_{\infty}\mathscr{D}\frac{\rho_{\infty}}{\rho_{Ce}}$. This yields a quadratic equation for a^{n+1} with two roots, only one of which is physical (i.e. $0 < a^{n+1} < a^{n}$ as expected for a burning particle). This physical root is used to obtain the new reaction radius at t^{n+1} and is given by

$$a^{n+1} = \frac{1}{2}\left[a^{n} - \frac{C\Delta t}{\delta} + \left(\left(\frac{C\Delta t}{\delta} - a^{n}\right)^{2} - 4C\Delta t\right)^{0.5}\right]. \tag{44}$$

The time required for a particle to fully react using the shedding model can be analytically computed as

$$\tau_r = \frac{1}{12i}\frac{\rho_{Ce}}{\rho_{D_2}}\frac{r_o^2}{DY_{\infty}}\left[1+cos\left(3arccos\left(\frac{1}{2}-\frac{\delta}{r_o}\right)\right)\right] + \frac{\delta^2}{iY_{\infty}D}\frac{\rho_{Ce}}{\rho_{D_2}}\left[\frac{r_o}{\delta}+ln\left(\frac{\delta}{r_o}\right)-1\right]. \tag{45}$$

This solution is obtained in two parts since the shedding model still uses the methodology for the growing shell model during the "transient" time when $a(t) < r_o - \delta$. The first term is obtained by first using the analytic solution to Eq. 27 (the solution for the variable outer radius model) to get the time needed for the reaction radius to go from $a = r_o$ to $a = r_o - \delta$. From there, the remaining reaction time (i.e. the second term on the left hand side) is obtained by integrating Eq. 27 with the modification that $b = a + \delta$ with constant δ rather than using a constant b. This leads to an implicit analytic solution form which can be solved by using the initial condition that $a(t = 0) = r_o - \delta$ and then solving for $a(\tau_r) = 0$. It should be noted that for small values of δ the "transient" time is small enough to be nearly negligible and the solution given by performing the integration of Eqn. 27 using $b = a + \delta$ with constant δ for the full reaction time can essentially be used as the verification test. Differences between the analytic solution and the numerical solution have been compared for a sampling of initial ejecta diameters and assumed shell thicknesses and have been found to be below 1.0%.

B. Particle Internal Energy Integration

1. Growing Shell Model

In order to integrate the energy equation (Eq. 30), the mass conversion rate $\dot{m}_{Ce}$ in Eq. 29 must be a real number for all values of $a \in [0,b]$. The initial reaction radius is $a = b$, which gives an infinite value at the first timestep. In order to resolve this issue, the particle properties are approximated by half-step values (i.e. $t^{n+1/2} = t^n + \Delta t/2$). This approach prevents the potential division by zero possibility. For the convective heat transfer term, it is best to integrate this term implicitly to avoid creating a timestep restriction.

If the time derivative of the particle's internal energy e_p is discretized with a first order explicit Euler approach the discretized energy equation becomes

$$\frac{\Delta e_p}{\Delta t} = \frac{Nu^n}{2}\frac{c^m}{\tau_T^n}(T_\infty^n - T_p^{n+1}) + \frac{\dot{m}_{Ce}^{n+1/2} h_r}{m_p}, \tag{46}$$

where the Nusselt number Nu, thermal relaxation time τ_T and the gas temperature T_∞^n are evaluated at t^n. The particle internal energy is related to the particle temperature by the particle specific heat so that $e^{n+1} = c^m T_p^{n+1} = e^n + \Delta e_p$. Combining these gives

$$\frac{\Delta e_p}{\Delta t}\left(1 + \frac{\Delta t}{\tau_T^n}\frac{Nu^n}{2}\right) = \frac{Nu^n}{2}\frac{c^m}{\tau_T^n}(T_\infty - T_p^n) + \frac{\dot{m}_{Ce}^{n+1/2} h_r}{m_p}, \tag{47}$$

where the change in internal energy for a particle Δe_p can be evaluated and used to update the internal energy at $e_p^{n+1} = e_p^n + \Delta e_p$.

The average mass conversion rate $\dot{m}_{Ce}^{n+1/2}$ is evaluated using Eq. 29

$$\dot{m}_{Ce}^{n+1/2} = i4\pi a^{n+1/2} b \mathscr{D} \frac{\rho_\infty Y_\infty}{b - a^{n+1/2}}, \tag{48}$$

where $a^{n+1/2} = a^n + b\Delta r_n^{n+1/2}$ and Eq. 39 can be evaluated at a half step to obtain

$$\Delta r_n^{n+1/2} = \cos\left(\frac{2\pi}{3} - \theta^{n+1/2}\right) - \cos\left(\frac{2\pi}{3} - \theta^n\right) \tag{49}$$

with

$$\theta^{n+1/2} = \frac{1}{3}\arccos\left(2\left[\frac{t^n + \Delta t/2}{\tau_r^n}\right] - 1\right). \tag{50}$$

Again, the possibility to divide by zero when $a = b$ is avoided by using the mass conversion rate $\dot{m}_{Ce}$ at a half-step.

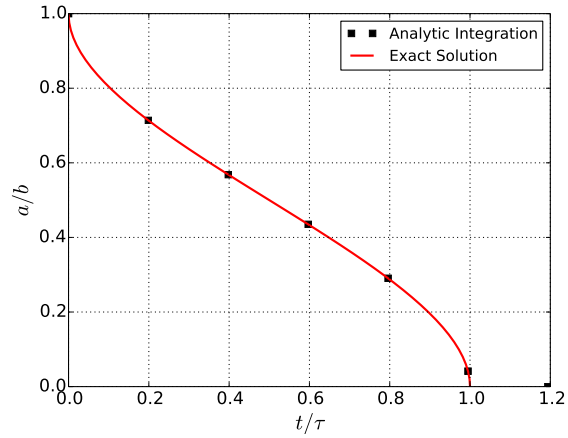


FIG. 5. Comparison of the discrete solution with only 5 timesteps to the exact solution discussed above.

Note that the process of obtaining the mass conversion rate at $t^{n+1/2}$ requires the reaction radius to be computed at that time as well. Thus, two timesteps are performed for the reaction radius during each interval Δt. The final reaction radius is obtained using $a^{n+1} = a^{n+1/2} + b\Delta r^{n+1}_{n+1/2}$, where

$$\Delta r^{n+1}_{n+1/2} = \cos\left(\frac{2\pi}{3} - \theta^{n+1}\right) - \cos\left(\frac{2\pi}{3} - \theta^{n+1/2}\right), \tag{51}$$

$\theta^{n+1/2}$ is given in Eq. 50 and

$$\theta^{n+1} = \frac{1}{3}\arccos\left(2\left[\frac{t^{n+1/2} + \Delta t/2}{\tau_r^n}\right] - 1\right). \tag{52}$$

Figure 5 compares the discretized solution with the exact solution when only 5 timesteps are used to resolve the reaction time. Even though an analytical solution is used for the reaction radius, it is still possible for the reaction radius to become negative. Thus, after each iteration the reaction radius is limited by the maximum of its value and zero, i.e. $a = \max(a, 0)$. Additionally, once the reaction radius is equal to zero it is not necessary to perform the reaction radius integration outlined above. Thus, the reaction step is only performed when $a > 0$. The evaluation of the reaction radius at $t^{n+1/2}$ and the above limiting steps are also performed if the shedding model is active.

2. *Phase Change Solution for the Shedding Model*

The shedding model's phase change solution is not handled via direct numerical integration of Eq. 26 (or its analog written in terms of particle internal energy). Instead, it operates by splitting

the particle energy update into two steps, one for the convective and reactive terms and one for the phase change term.

The convective and reactive terms are handled as described above with no change. If an ejecta particle reaches the condition for phase change, then the phase change energy term is handled after the convective and reactive terms as follows. At the onset of a phase change, the total energy requirement for phase change is computed as

$$H_{fus}^{n} = \frac{4}{3}\pi h_{fus}^{o}\rho_{CeD_2}((b^n)^3 - (a^n)^3)\,. \tag{53}$$

In subsequent time steps, a second evolution equation for H_{fus} is solved using an explicit time marching scheme in sync with the solution of Eq. 30 which states that

$$H_{fus}^{n+1} = H_{fus}^{n} + \left(\frac{dH_{fus}}{dt}\right)^{n}\Delta t - m_p(\Delta e_p)_{c+r} \tag{54}$$

where

$$\frac{dH_{fus}}{dt} = 4\pi h_{fus}^{o}\rho_{CeD_2}(b^2 - a^2)\frac{da}{dt} \tag{55}$$

and $(\Delta e_p)_{c+r}$ is obtained from solving only the convective and reactive portions of the energy equation. The above scheme can be seen as essentially a re-writing of Eq. 26 under the assumption that $\Delta e_p = 0$. These equations are solved in sync until $H_{fus} \leq 0$, at which point the phase change is assumed to be complete and the energy solution scheme reverts back to the growing shell model approach.

III. RESULTS

A variety of limiting cases were considered to ensure the equations were coded correctly. The first includes verifying that non-reacting particle temperatures asymptote to the surrounding gas temperature on a timescale on the order of the thermal relaxation time τ_T. This ensures that the heat transfer model works correctly independent from any reactions. The second test focuses on the diffusion controlled reaction and ensures that a particle with diffusion controlled reactions reacts completely. There are two different solutions to the reaction equation for the shedding and non-shedding models.

A. Simulation Configuration Overview

The multiphysics simulations used to analyze the performance of the hydriding models in a full-scale context are simulations of two sets of experiments performed by Buttler *et al.* in 2017 and 2019. These experiments looked at the transport of cerium ejecta, both solid (2017) and liquid (2019), in vacuum as well as in helium and deuterium gas in order to study the differences between inert and reactive transport. The layout of the experiments is described in detail in [31], but an overview is provided here for clarity before jumping into the simulation setup.

Figure 6 shows the general configuration for the ejecta experiments under consideration. A 50 mm diameter cylindrical metal target (in this case cerium) which has perturbations etched onto its back side is placed flush on one end of a containment canister. The unperturbed side of the target is placed next to a high explosive. This high explosive is detonated and drives a shock through the target, producing ejecta particles off the perturbed finish after shock breakout. The ejecta form a cloud and travel into the ambient environment set inside the remainder of the vessel which can be a vacuum or loaded with a gas (helium or deuterium) before the shot. This ambient environment contains various diagnostic tools which record data from the transporting ejecta. The most relevant of these for this report are lithium-niobate (LN) pins to record accumulated ejecta mass, laser-Doppler velocimetry (LDV) probes to record ejecta velocity distributions, and infrared (IR) cameras to record the observed radiance temperatures in the ejecta cloud. The LN pins capture mass readings at a fixed height above the initial target location. The LDV probes are pointed perpendicular to the direction of motion of the ejecta cloud and capture data throughout the experiment while the infrared cameras are set to record data at fixed heights perpendicular to the direction of motion of the cloud and capture data only for the time which the cloud is in the field of view of the cameras.

A schematic of the simulation domain used to replicate the experiments is shown in Fig. 7. The experiments are simulated using a two-dimensional, axisymetric framework, with the axis of symmetry taken to be the center of the metal targets, in order to reduce computational costs by only requiring half of the experimental geometry to be simulated. Instead of simulating the detonation of the high explosives which are used in the experiments, a flyer plate of the same material as the target is used to initialize the ejecta-producing shock. The flyer plate is placed at the bottom of the simulation domain, directly below the target, and is initialized with an axial velocity equal to that of the reported jump velocity for the particular experiment being simulated. The 2017

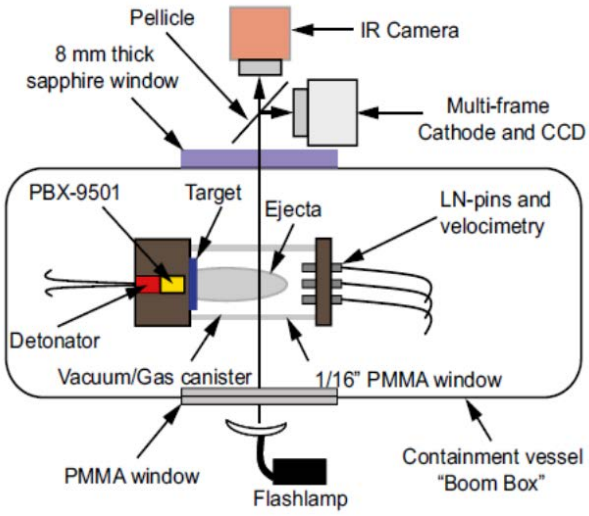


FIG. 6. A general schematic showing the configuration of the experimental setup for Buttler's 2017 and 2019 experiments taken from [32]. The location and number of the IR cameras is different in the experiments simulated for this report but the remaining setup is qualitatively correct.

experiments reported jump velocities between 1860-1950 $\frac{m}{s}$ while the 2019 experiments reported jump velocities between 2280-2340 $\frac{m}{s}$. The decision to use a flyer plate rather then simulating the detonator and high explosive combination used in the experiments was made in order to reduce the multiphysics complexity of the simulations. At the same time, this decision also reduces their computational cost since a much higher mesh resolution would be required in the high explosive region in order to resolve the propagating detonation wave. As this work is primarily focused on modeling the ejecta transport seen in the experiments, this decision is justified so long as the flyer plate setup generates similar shock jump conditions to those reported by Buttler *et al.* [31] and also generates the proper amount of ejected mass, similar velocity distribution, size and initial particle temperature as seen in a given experimental shot. Above the flyer plate and target combination, the remaining simulation domain is initialized as quiescent gas or vacuum at the conditions reported in Tables 1 and 3 in [31].

The simulation diagnostics used to capture data to compare with the experiments are also initialized into this domain. To compare with the LDV data, ejecta velocities are recorded along the axial direction of the simulations using two beams of radius 1.2 mm with origins at the top of the simulation domain, spaced 3.8 mm apart, pointing towards the target. To compare with the IR camera data, ejecta temperatures are recorded in the radial direction from a beam situated a height

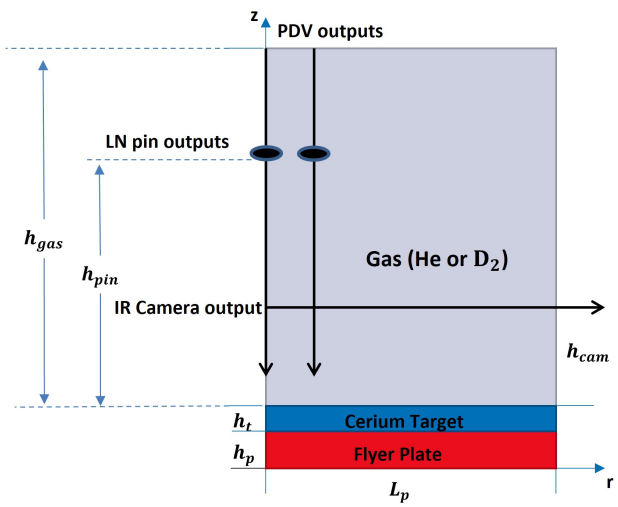


FIG. 7. Schematic showing the configuration for simulations of Buttler's 2017 and 2019 experiments.

Dimension	2017 Value	2019 Value
L_p	2.500 cm	2.500 cm
h_p	2.000 cm	2.000 cm
h_t	3.000 mm	1.750 mm
h_{cam}	1.000 cm	1.025 cm
h_{pin}	2.200 cm	2.325 cm
h_{gas}	3.965 cm	4.065 cm

TABLE I. Dimensions of geometry used in simulations of Buttler experiments.

h_{cam} from the initial target surface and then mass averaged. Finally, in order to compare with the LN-pin data, accumulated ejecta masses are recorded on two, 0.64 mm radius "foils" which are located a height of h_{pin} from the initial target surface and spaced out equally to the velocity diagnostics. A main discrepancy between the experimental LDV and IR camera diagnostics and their simulation counterparts must be noted. The simulation diagnostics are capable of recording the ejecta velocities and temperatures along the entirety of the "beam" they are set to track. However, the experiment diagnostics will have their field-of-view along those directions optically blocked after a certain distance due to the optical thickness of the travelling ejecta cloud. This discrepancy can (and most likely does) lead to additional uncertainty in the simulation data and needs to be accounted for when performing a comparison of the two data sets.

The models discussed in Section I were implemented in the Lagrangian hydrocode FLAG [41],

Quantity	Symbol	Value	
Metal specific heat	c^m	37.7	J/kg-K
Particle density	ρ_p	6680	kg/m^3
CeD_2 Heat of formation	h_f^0	210E+6	J/kmol
Solid CeD_2 Diffusion coefficient	$\mathscr{D}$	4.0E-4	cm^2/s
Particle atomic mass	W_p	140	g/mol
Gas atomic mass	W_c	4.0	g/mol
Gas specific heat ratio	γ	1.4	

TABLE II. Model parameters used for cerium-deuterium simulations.

which was also used to perform the simulations described above. A more detailed explanation of the governing equations and pre-existing ejecta models of FLAG which are relevant to this work are presented in Appendix A.

Table II lists the model parameters that are common to both the solid and liquid cerium cases such as material properties, heat of formation, molecular mass, etc. The stoichiometric mass ratio (Eq. 7) is $i = 35$ and the heat of reaction $h_r = h_f^0/W_p$ = 1.5E+6 J/kg. Most of the metal properties were obtained from [42] and the gas properties were obtained from [43]. The heat of formation of cerium dihydride was obtained via calculations performed by Dan Shepard and is consistent with the range of values documented in [38, 44].

The hydride diffusion coefficient was obtained via QMD simulations [45]. It is intended to give a rough, order of magnitude, estimate of the diffusion coefficient for the temperature range considered. In a discussion with Sheppard, it was suggested that we account for the change in diffusion coefficient when the hydride layer melts by increasing it by an order of magnitude. Thus, when the phase change model completes a melting/freezing process, it increases/decreases the value of $\mathscr{D}$ by a factor of 10.

For simulations of the 2019 cases in deuterium which use the shedding model, two additional material properties are required. The latent heat of fusion of cerium dihydride was taken to be 1.42 J/kg which is in reality that of lithium hydride. However, this value is taken to be a reasonable approximation and is readily found in the literature unlike that of cerium dihydride. Lastly, the melting temperature of cerium hydride is assumed to be 1690 K, which is taken as a lower-bound estimate from the work done by Schulze [46] where the melt temperature was estimated to be

TABLE III. Individual input parameters used for the 4 and 8 atm cases.

Chamber Material	vacuum	He	He	D_2	D_2
Initial Gas Pressure [atm]	10^{-4}	2.03	9.45	4.05	8.11
Shocked gas temperature T_∞ [K]	-	1030	1030	720	740
Shocked gas density ρ_∞ [kg/m^3]	10^{-3}	0.948	4.38	2.3	4.5
Free surface velocity u_j [mm/μs]	1.95	1.91	1.88	1.91	1.86

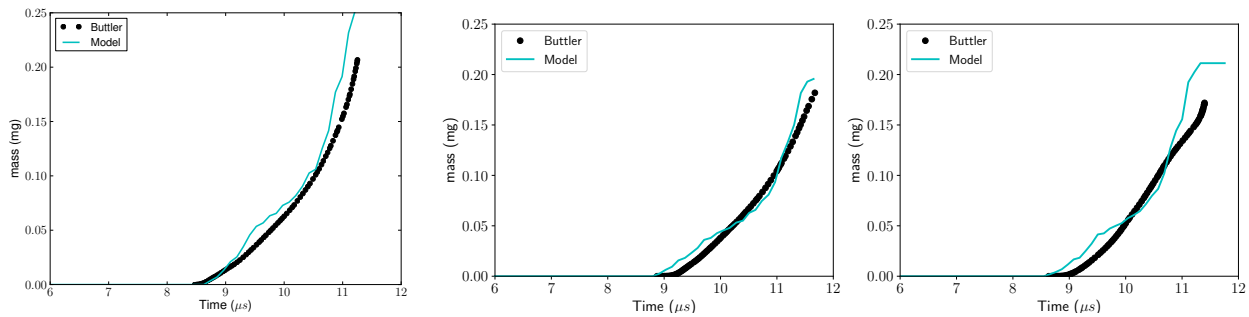
roughly 1700 K.

B. Growing shell model

The accuracy of the non-shedding model is evaluated by simulating the 2017 solid cerium ejecta experiments documented in [30–32] and briefly described in Section III A. These experiments analyzed both tin and cerium metals, but we focus solely on the experiments using cerium metal ejecta. The experiments send a shock wave through a cerium plate that is machined so that ejecta particles are created with a mean diameter of $13\mu m$. Table III contains the input parameters for each tested condition. Each experiment was performed with the same amount of explosive, which produced a shocked metal temperature $T_p = 990$K. The mean particle diameter for each of these shots was observed to be 13μm.

1. Source model calibration

The source model used to model the creation of subgrid-scale ejecta particles is created for materials that have been shocked hard enough to melt. In the 2017 experiments, the cerium metal is still solid upon release of the shock wave. An immediate question arises; how do we create a cloud of solid ejecta particles when we don't have a solid ejecta source model? Recall that the 2017 solid experiments were originally designed to create liquid ejecta, but the equation of state used to design the experiments did not accurately predict the melt temperature, so the experiments got close to creating liquid ejecta, but not close enough. It was for this reason that we thought that the liquid ejecta model might do a reasonable job at predicting the production of solid ejecta. Fortunately, we found that the liquid ejecta source model did a reasonable job predicting the mass production, velocity, size and initial temperature of the ejecta cloud with some minor calibrations.



a) Vacuum b) Helium c) Deuterium

FIG. 8. Mass produced via the RMI source model with SSVD distribution for vacuum, helium, and deuterium environments.

The solid ejecta are created using the unmodified self-similar velocity distribution (SSVD) model because it produces ejecta at a slower rate. Also, the perturbation wavelength is artificially enlarged from the 100μm reported from the experiemnts to 166μm and the ejecta production time was limited to 0.4μs. This combination provides the best match of ejecta production rate while simultaneously giving a mean particle diameter of 12μm. Figure 8 shows how well the mass ejected over time from the simulation matches the experiments.

It should be noted that the time denoted as the zero-time (i.e. $t = 0$) is not consistent between the experiments and the simulations. Thus, in the LN-pin results shown in this report, the experimental data is time shifted such that the time of arrival of the ejecta clouds to the pins is matched between the two. This allows for the isolation of the total accumulated ejecta mass and the mass accumulation rate comparisons and analysis from any effects from errors in judging the zero-time and its uncertainties from the experiments.

C. Temperature calibration

The radiance measurements for helium shots Ce2 and Ce6 in [31] are roughly constant at 900K. The hydrodynamics simulations show the post-shock gas temperature to be 1030K. In order for the particle radiance measurements to remain constant, the particles should be in thermal equilibrium with the gas. The data in Fig. 4 of [31] show that the radiance measurement decreases in the 2atm condition and increases slightly in the 9atm condition. Since the post-shock gas has uniform properties not including the effects of the ejecta cloud, one can conclude that the particle temperatures are in thermal equilibrium with the post-shock gas during the measurement time.

It is possible then to obtain an estimate on the emissivity of these solid cerium ejecta particles. Buttler *et al.* [31] report a temperature exponent appropriate for the infrared cameras used is 3.2. Thus, we can estimate the emissivity to be

$$\varepsilon = \left(\frac{R_p}{T_p}\right)^{3.2}, \tag{56}$$

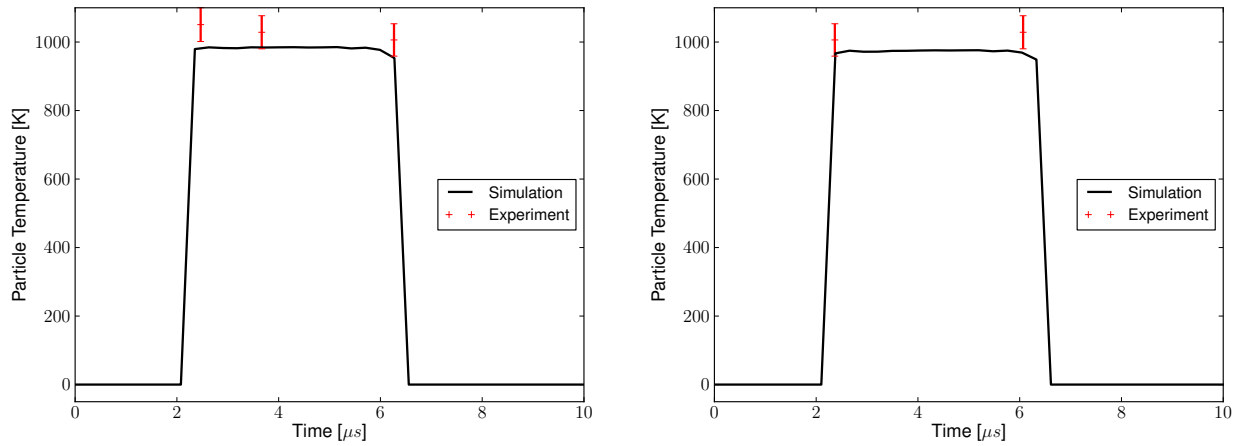
where R_p is the particle radiance measurement and T_p is the particle temperature. If we assume the particle radiance averages 900K and the equilibrium temperature is 1030K, then an emissivity of $\varepsilon = 0.7$ is obtained.

Figure 9 compares the particle temperature predicted using the simulation and the experimentally observed particle temperatures calculated from the data using the emissivity value $\varepsilon = 0.7$ for cerium ejecting into helium. Error bars are also obtained for the experimentally measured temperatures by differentiating Eq. 56 using the chain rule to obtain the differential for the particle temperature in terms of the radiance quantities as:

$$\delta T_p = \frac{dT_p}{d\varepsilon}\delta\varepsilon + \frac{dT_p}{dR_p}\delta R_p = R_p \varepsilon^{-\frac{1}{3.2}} \left[\frac{\delta R_p}{R_p} - \frac{1}{3.2}\frac{\delta\varepsilon}{\varepsilon}\right] \tag{57}$$

where $\frac{\delta R_p}{R_p}$ is the uncertainty in the experimental radiance measurement (given as 2% in [31]) and $\delta\varepsilon$ is the uncertainty in the value of the emissivity (which is assumed to be 0.15 in this work). Similar to the experimentally measured particles, the predicted particle temperatures from the simulation are roughly constant with time as well. As discussed above, the temperature from the simulation is measured from the side using a numerical averaging diagnostic so that a similar type of measurement is made between experiment and simulation. The abrupt rise and fall of the particle temperature corresponds to the emergence of the particle cloud into and then out of the diagnostics field of view.

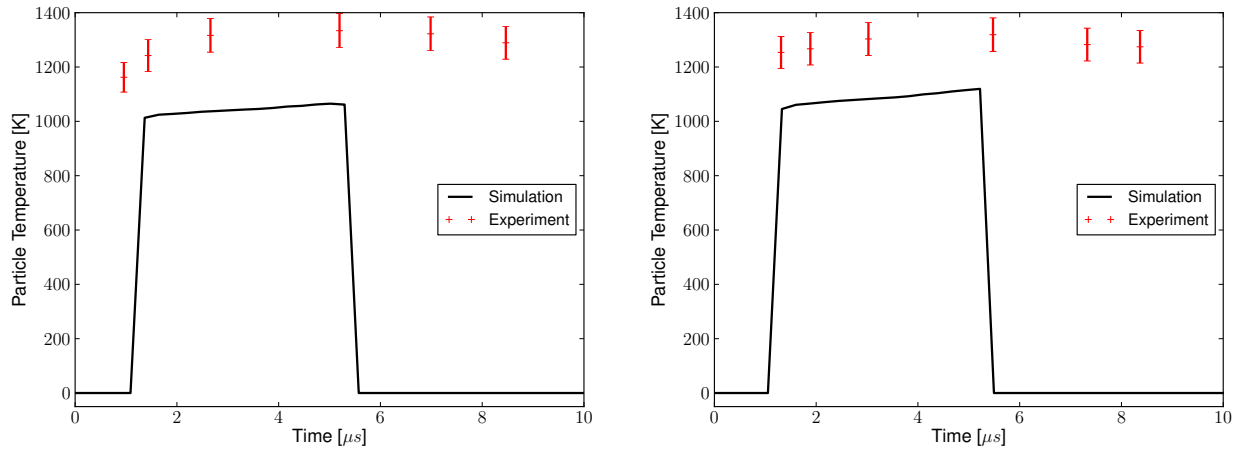
Figure 10 shows the particle temperatures obtained from the simulation via the temperature diagnostic and the experimentally measured particle temperatures after converting from radiance. The experimentally observed particle temperatures show a rise over about 5μs followed by a subsequent decline. The model particle temperature is 200-300K lower than the experiment and almost within the experimental error bars, but still rises, indicating that the heat release from reaction exceeds the convective heat loss to the gas. However, the abrupt drop in temperature to $0K$ just before 6μs on the plot indicates that there are no more ejecta particles to be observed for measurement. It is not yet clear why the simulation is missing these late time particles that the experiment appears to capture. It is possible that this is an outcome of certain shortcomings in



a) 2 atm

b) 9.45 atm

FIG. 9. Temperature measured by simulation compared with experiment for the 2atm (left) and 9atm (right) He gas conditions.



a) 4 atm

b) 8 atm

FIG. 10. Temperature measured by simulation compared with experiment for the 4atm (left) and 8atm (right) deuterium gas conditions.

the ejecta sourcing model used in the solid reactive simulations, such as the previously mentioned molten target assumption or the lack of accounting for any chemical reactions on the growing metal spikes during the RMI evolution assumed by the model.

TABLE IV. Individual input parameters used for the simulated liquid cerium cases.

Chamber Material	vacuum	He	D_2
Initial Gas Pressure [atm]	10^{-3}	5.32	4.01
Shocked gas temperature T_∞ [K]	-	1280	945
Shocked gas density ρ_∞ [kg/m^3]	10^{-2}	2.70	2.73
Free surface velocity u_j [mm/μs]	2.34	2.28	2.33

D. Shedding Model

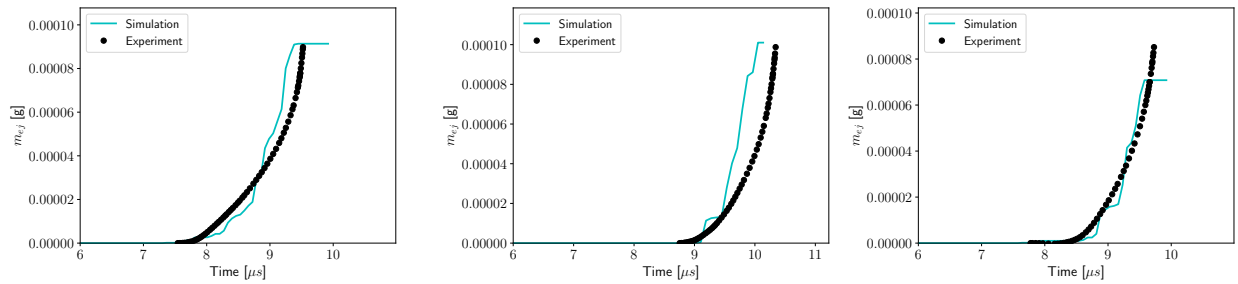
The performance of the shedding model is evaluated by simulating a subset of the 2019 liquid cerium ejecta experiments documented in [31]. These experiments focused only on cerium as the target metal with the setup referenced in Section III A. The main difference to the solid ejecta experiments is the cerium plates used in these experiments were thinner and twice the amount of high explosive was utilized in order to produce liquid ejecta. Additionally, the targets were machined such that the resulting ejecta have a mean diameter of roughly 2 μm. Table IV contains the input parameters for each simulated experimental condition. The liquid experiments of interest to this work were performed with the same quantity of high explosive driving the shock through the cerium plate for all gas environments, producing a shocked metal temperature of $T_p \approx 1070K$ which is the melting point for pure cerium.

All simulations involving cerium being ejected into a deuterium environment are performed with both the shedding and shell phase change models active unless otherwise noted. Simulations of cerium ejecting into other environments are run using the variable (growing) outer radius reaction model.

1. Ejected Mass Results

Unlike the simulations of the solid cerium ejecta experiments described in Section III B, the simulations of the liquid cerium ejecta produce a molten target surface upon release. Therefore, no source model calibration was required. Figure 11 shows comparisons between the experimentally measured accumulated ejecta mass on the deployed LN pins and the measured mass accumulation from the hydrocode diagnostic.

The results show that the proper amount of ejecta mass is generated for the inert environment



a) Vacuum b) Helium c) Deuterium

FIG. 11. Mass produced via the RMI source model for vacuum, helium, and deuterium environments for liquid ejecta

simulations while the deuterium environment (i.e. the one which utilizes the shedding model) appears to underestimate the ejected mass. However, this may be due to a shortcoming in the current state of the hydrocode diagnostic in that the diagnostic can only record accumulated mass from any particles which travel through it. In its current form, the diagnostic has no method to track the mass of any shed hydride which passes through it. In Figure 12, we show a comparison of the ejected mass output for simulations in deuterium with either the growing shell or the shedding model active. This plot demonstrates that the correct mass of cerium is also being ejected in the deuterium environment. The described shortcoming in the hydrocode mass accumulation diagnostic will be resolved in future work.

Another point of interest is the apparent "stepping" that can be seen in the simulation mass outputs. There are two reasons that this appears in the plots. The first is a numerical artifact from the hydrocode in that the simulation reports the total ejecta mass which has hit the pin diagnostic at a given time. Thus, it is prone to scenarios where multiple time steps can occur where no ejecta packets hit the area defined as the pin surface and then one step occurs where a noticeable ejecta mass crosses the plane. This phenomenon will appear as a "step" on the plotted result. The second is coupled with the first and relates to both the location of the pin diagnostics in the simulation and how the hydrocode numerically sources the ejecta. The pin diagnostics are clustered around the centerline of the cylindrical target in the experiments, so near the axis of symmetry used in the simulations. The axis of symmetry will also be the region in the simulations with the lowest amount of expected ejected mass. When the hydrocode sources ejecta, the RMI model gives the amount of ejecta mass to produce at the sourcing face and then combines that with a constant (user-defined) loading factor for how many physical particles are represented by a single computational

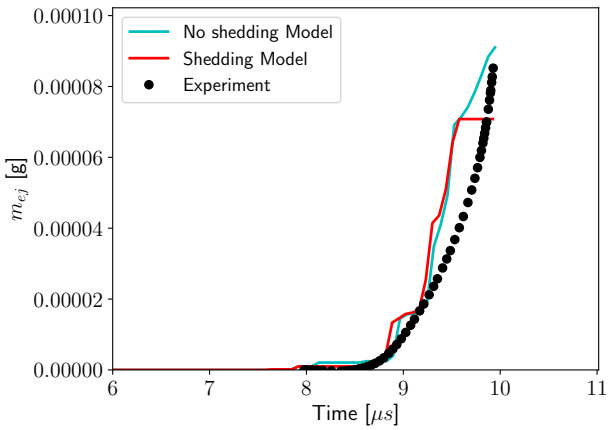


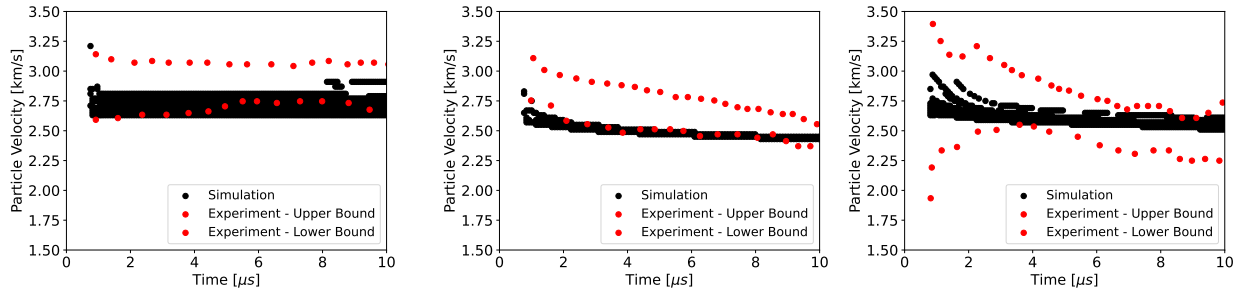
FIG. 12. Comparison of the mass production recorded in simulations by the hydrocode diagnostic in deuterium environments with the shedding model active and inactive. The active model simulation is run using a shell thickness of 20 *nm*.

particle. With this logic, fewer computational particles will be created near the axis of symmetry compared to regions radially further from the axis and this fact can potentially exacerbate the numerical artifact described above.

The three cases produced ejecta with average diameters between 1.7-2.3 μm which falls in line with the experiments which reported ejecta diameters between 1.8-2.4 μm. Of note is that, while the simulations did produce the correct amount of total ejected mass, their agreement to the LN pin data as a time history is not a total match. However, the simulations do produce mass accumulation curves which appear to have a similar late-time slope to the experimental curves. This outcome points to the possibility of using the mass accumulation rate on the pins as a better validation metric than the raw mass accumulation in future studies.

2. *Ejecta Velocity Results*

Figure 13 compares the simulated ejecta velocities along the experimental LDV lines of sight to the bounds of the experimental LDV data. The vacuum case shows good agreement with the experimental data in that the simulated ejecta velocities fall within the LDV data bounds and also remain fairly constant throughout the simulation. This result demonstrates that the ejecta sourcing model is initializing the ejecta velocities in the same magnitude range as seen in the experiments when ambient fluid effects are not taken into account even if the exact velocity distribution to use is under question. Likewise, the helium case also has ejecta velocities which fall within the LDV



a) Vacuum b) Helium c) Deuterium

FIG. 13. Simulated ejecta velocities in vacuum (left), helium (center) and deuterium (right) conditions plotted against the upper and lower bounds of the Buttler experiment ejecta cloud LDV.

data bounds and show a similar deceleration slope to the experiments. However, these velocities do not fill the entirety of the experimental bounds and tend to the lower velocities after initial ejecta creation. Since the ejecta velocities tend to the lower bounds of the experimental data even on creation, this points to a possible small flaw in the ejecta sourcing model's velocity component rather than the hydrodynamic drag model being used. Even so, these results show promise in the ability of the current ejecta models to handle ejecta transport in the absence of reactions.

The deuterium simulation also shows ejecta velocities which fall within the LDV data bounds and seem to have a bulk deceleration curve which agrees well with the experiments for times less than 5 μs. Additionally, at times less than 2 μs, a small-scale non-monotonicity in the ejecta acceleration can be seen with a larger-scale version appearing at times closer to the end of the simulation. This is important as this observed non-monotonicity at early to moderate times in the experiments is what pointed to a possible break-up mechanism occurring in the reactive, liquid ejecta cases. One possible reason why the acceleration behavior is different, or at least appears to be different, in the simulations could also be due to the velocity diagnostic method.

The working theory behind a break-up mechanism being responsible for the observed non-monotonic accelerations in the experiments is that the LDV probe would see the leading edge of the ejecta cloud, which moves at a velocity faster than the post-shock fluid velocity, first. As those ejecta reacted and eventually broke up (or flaked shell material off), their size would shrink and they would equilibrate to the slower, ambient gas velocity more quickly. This would cause them to recede into the cloud as the larger ejecta behind them, which had not fully reacted/broken up, sped past them as their larger inertia allowed them to remain faster than the gas for a longer time. This cycle would repeat for most of the ejecta cloud's transport, causing the observed, step-like

behavior in the velocimetry data extracted from the experiment.

One major issue with attempting to capture this full scale phenomenon in the simulations, if the working theory is correct, exists. The above description relies on the existence of a limit to the optical depth of the LDV probe, otherwise the probe would see all of the ejecta down its line of sight regardless of size. As mentioned quickly in Section III A, the simulation diagnostic records the ejecta velocities along the entire line of sight of the beam. Thus, it does capture all of the ejecta at once, regardless of which specific particles are leading the cloud. Due to this, it is not surprising that the simulation diagnostic being shown in Figure 13 does not capture this phenomenon in full. A better comparison would be to look at the velocity and size contours in the cloud as a whole to determine if the leading edge of the cloud oscillates in both diameter and velocity. However, no current experimental data exists which can be easily compared to that analysis. So, at best, this approach would serve only to either uphold or deny the feasibility of the working theory of break-up being responsible for the non-monotonic acceleration.

3. Ejecta Temperature Results

For the analysis of ejecta temperatures from the liquid experiments, the earlier calibrated emissivity values are not used since analysis performed in [31] from the liquid ejecta experiments yields an emissivity estimate of $\varepsilon = 0.6$. The radiance camera appropriate temperature exponent of 3.2 will still be used. Therefore, the reported ejecta temperatures from the liquid experiments are transformed from radiance temperatures to physical temperatures via the expression $T_p^{3.2} = \frac{1}{\varepsilon} R_p^{3.2}$.

The data taken from the liquid ejecta experiments show a clear difference in behavior between the inert and reactive ejecta thermal behavior. The radiance measurements for the 2019 shots into helium exhibit a slow, but steady rise in temperature at early times from their initial temperature of roughly 1070 K (the melting temperature of cerium) up to an near equilibrium temperature of roughly 1280 K which is the reported post-shock temperature of the helium. This shows that the ejecta in these experiments were still undergoing a thermal equilibration to heat up to the ambient gas temperature during the time the IR camera data were taken. In contrast, the shots into deuterium show a quick rise in the ejecta temperatures from their breakout temperature of roughly 1070 K due to the occurring exothermic hydriding reaction, similar to that seen in the solid 2017 ejecta shots seen in Fig. 10. However, unlike the cooling seen in the solid experiments, the liquid radiance temperature is then followed by a plateauing temperature of around 1400 K at moderate

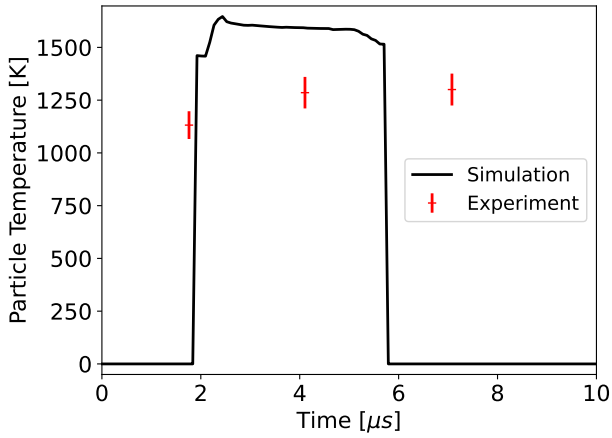
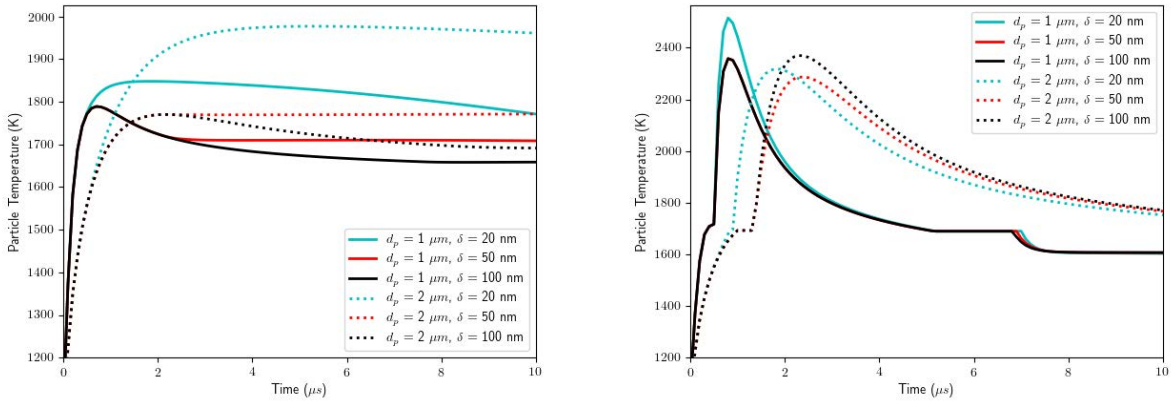


FIG. 14. Simulated mass-average particle temperatures compared to the experiment in helium conditions.

to later times (approximately 2 μs post-shock breakout). This radiance temperature corresponds to a physical temperature of 1642 K when transformed as defined previously and is near the theorized melt temperature of cerium dihydride (1700 K). The stalling of the ejecta temperatures near this postulated melt point led to an initial working theory that the cerium hydride shells were undergoing a melting process after an initial transient transport period. This temperature plateau is the main difference between the thermal behavior of the solid and liquid cerium ejecta experiments in deuterium.

Figure 14 compares the ejecta temperatures from simulations of the 2019 experiments into helium with the radiance measurements converted into temperatures. The ejecta temperatures from the simulations are obtained from calculating a mass-average temperature from every ejecta particle which is within a cylindrical region parallel to the target surface, representing the IR camera's line of sight, at a given time. Separate calculations output the simulation ejecta temperatures using a surface area average and showed minimal differences to the mass averaging approach (specifically, slight decreases in recorded temperatures for the inert cases and slight increases for the not yet discussed reactive cases). Thus, for this work, we use the mass average temperatures for discussion.

Similar to the results seen from simulations in Section III B, these inert simulations also show two distinct jumps in temperature. The earlier-time jump is the initial appearance of the cloud in the diagnostic line-of-sight and the later-time jump is the cloud exiting the field of view. These simulations also suffer from the shorter time duration of the cloud in the diagnostic line of sight compared to the experiments. Unlike the helium experiments, the simulation appears to severely



a) No phase change model

b) With phase change model

FIG. 15. Ejecta temperatures for single cerium particle transport in deuterium using the shedding model without (left) and with (right) the phase change model for varying values of initial ejecta diameter and δ.

overestimate the initial ejecta temperatures. The simulations are still showing a thermal equilibration occurring at a similar rate to that of the experiments, but the simulations are cooling rather than the experimentally observed heating. The cause for the initial overestimation of the simulated ejecta in helium is still under investigation.

In order to do a proper temperature comparison between simulations of the liquid cerium experiments and the experiments themselves, the effect of the constant shell thickness parameter used in the shedding model, δ, must also be analyzed. Figure 15 shows the temperature evolution for single cerium particles transporting in deuterium at a velocity of 460 m/s (roughly the initial relative velocity of the fastest ejecta in the experiments) at various initial sizes and values of δ both with and without the shell phase change portion of the shedding model active. Both plots clearly show an effect on the temperature behavior from the chosen value of δ, regardless of initial ejecta size. Smaller values of δ tend to lead to higher peak temperatures in the particles. With the phase change model active, the amount of time needed to melt the shell also scales with δ, as expected. However, the most important item to note when using the phase change model is the much larger peak temperature values which arise when the phase change is active on the order of 400-500 K. The diffusion coefficient of deuterium through liquid cerium hydride increases by an order of magnitude compared to that of solid cerium hydride in the model (see Section I E). Since increasing the diffusion coefficient will in turn increase the rate of the occurring hydriding reaction, the phase change in the shell subsequently increases the heat released due to the reaction.

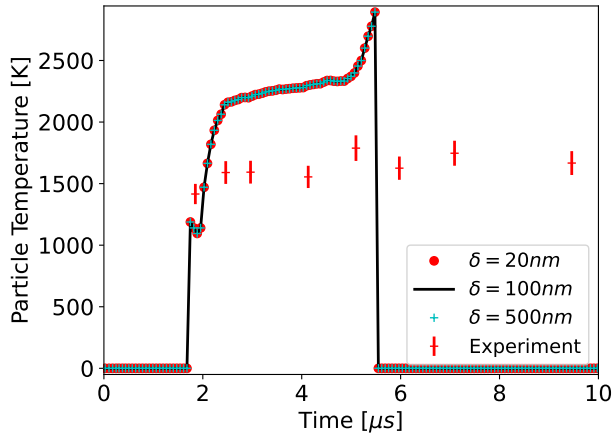


FIG. 16. Simulated mass-average particle temperatures compared to the experiment in deuterium conditions for various shell thickness values.

This pushes the peak temperature upwards so long as enough of the molten cerium core exists to drive the reaction and produce more heat than is lost to convection with the ambient gas. Thus, the temperature peak is more intense for smaller particles but is spread out over a longer period of time for larger particles, as demonstrated in Fig 15.

Figure 16 compares the particle temperatures predicted from the simulations for varying values of δ along with those recorded from the experiments. The results are somewhat surprising as it appears that varying the shell thickness has a minimal effect, if at all, on the mass-averaged temperatures recorded along the line of sight of the simulation diagnostic. It should be noted that this lack of effect of δ on the simulation output may be due to competing effects of the mass-averaging of the ejecta particle temperatures in the cloud and the ability of the simulation diagnostic to see through the entire simulation cloud masking one another. No conclusions should be made as to the effect of δ (or other model parameters) on the outputs which would arise from utilizing a simulation diagnostic which perfectly mimics the data output from the experimental diagnostic.

All three values of the assumed shell thickness show simulated temperatures which, again, appear to overestimate the initial ejecta temperatures from the experiments. However, instead of plateauing near the melt point for cerium dihydride, they continue increasing until violently spiking upwards towards temperatures above 2500 K before the cloud leaves the diagnostic beam. This is most likely not realistic and hinted that, possibly, the diffusion coefficient being used for D_2 through CeD_2 was too high, as too high a value would cause the hydriding reaction to go

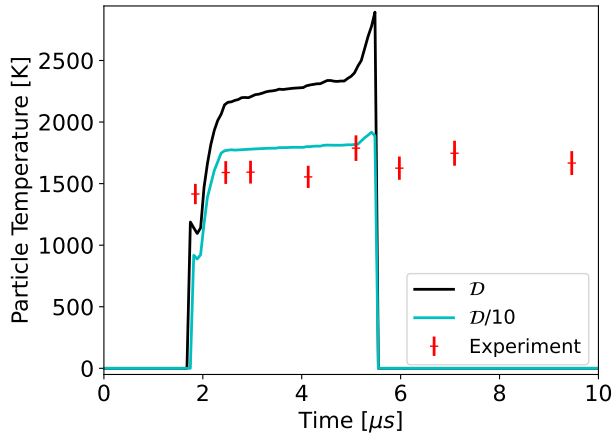


FIG. 17. Temperature measured by simulation compared with experiment using the original diffusion coefficient for Ce through CeD_2 and using a coefficient reduced by a factor of ten using a shell thickness of 100 nm.

unphysically fast and cause the temperature to unphysically spike in turn due to too much reactive heat release. This effect would also be exasperated by the fact that most of the hydride shells in the simulations are melted (or in the process of melting) since the average ejecta temperature exceeds the given melt point of the shell in the model (1690 K), thus leading to the coefficient jumping up by an order of magnitude once melting completes.

Since the diffusion coefficient is a rough estimate and the model particle temperatures are so much higher than the experimentally measured temperatures, an additional simulation was run with the diffusion coefficient reduced by a factor of 10. Figure 17 shows the effect of reducing the diffusion coefficient by a factor of ten on the simulated ejecta temperatures. The two plotted simulation curves can be viewed as a form of uncertainty propagation of the uncertainty in the value of $\mathscr{D}$ to the output uncertainty in the simulated ejecta temperatures. The modified diffusion coefficient simulation still slightly overestimates the ejecta temperatures from the experiment, but the difference is significantly improved with the reduced diffusion coefficient. The simulated average never exceeds 2000 K and the slope of the temperature time history falls much more in line with the experiment. Also, reducing the diffusion coefficient eliminates the violent spike seen in the simulated temperatures. However, as stated in Section III B, there are a multitude of other uncertainties in the experimental data and the diagnostic used to extract the temperatures from the simulations which are also playing a substantial role in the temperature mismatch, as one still exists even with the reduction of the factor to the lower bound of its uncertainty.

The results shown have additional uncertainties placed on them due to the utilization of the shedding model as the model is still missing important physics incorporating shell fracture, ejecta volume fraction and gas diffusion, among other physics. An important unknown of interest is adsorption, or the limit on the mass fraction of deuterium gas that can be seen at the surface of the hydride shell. This specific value for D_2 diffusing through CeD_2 is not incorporated in to the shedding model and will also have a direct impact on the diffusion rate if implemented. As seen in Equations 21 and 26, the value of the concentration of hydrogen/deuterium at the shell surface is a linear factor in the governing ODEs for the reaction front radius and the particle temperatures similar to the diffusion coefficient. Thus, it can be estimated that altering the value of Y_∞ by a given factor would have a similar impact on the ejecta temperature evolution as altering $\mathscr{D}$ by the same factor. More importantly, many of the material properties for CeD_2 are also unknown so the model, in its current form, uses approximations for many of them. Future work will be needed to obtain better approximations, if not exact values, for these properties as well as to incorporate the more advanced physics aspects into the framework.

IV. CONCLUSIONS

A simple model is developed to capture the exothermic hydride reaction of cerium material ejected into a deuterium gas. The model assumes the particle remains spherical, is hydrodynamically stable and does not breakup beyond its initial size. The hydride reaction is controlled by diffusion of deuterium through the hydride layer and reacts infinitely fast at the pure cerium metal surface. The model further imposes that, for small ejecta sizes, the developing hydride layer flakes at its outer edges creating nano-scale particles which immediately equilibrate with the carrier gas. The model is able to increase the particle temperature above the initial formation temperature and the surrounding gas temperature. However, the model particle temperatures are systematically lower than the experimentally observed temperatures for solid ejecta and are systematically higher than the experimentally observed temperatures for liquid ejecta. Furthermore, the source model is unable to produce the late time ejecta observed in the experiments. It should be noted that the source model was not altered for any reaction effects. The inclusion of these effects may help with the under/overprediction of the particle temperatures in the solid/liquid ejecta cases.

There is significant uncertainty in the diffusion coefficient for deuterium into the hydride layer. A separate simulation with a diffusion coefficient an order of magnitude smaller than the value

used in all other simulations produces particle temperatures that are almost within the experimental error bars. However, this reduction in diffusion coefficient will likely make the model under-prediction of the particle temperature of the solid ejecta experiments even worse. These inconsistencies are a bit puzzling and given the number of assumptions made and measurement of an entire cloud, rather than individual particles, makes identifying a cause difficult. However, it is possible that adsorption effects of deuterium onto the particle surface, an effect not included in the current model, could possibly help with the temperature inconsistency by limiting the diffusion rate through saturation. New experiments capable of measuring the temperature of a single particle relative to a gas of known temperature while visualizing the hydride crust would help identify weaknesses in the model and provide a better means of validation.

V. ACKNOWLEDGEMENTS

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Appendices

A. GOVERNING EQUATIONS AND EJECTA MODELS

The FLAG hydrocode handles the carrier fluid and bulk solids in a Lagrangian manner using a staggered mesh approach where mesh vertices are Lagrangian points where velocities are stored and zone centers store the remaining thermodynamic variables such as density, pressure, temperature and specific internal energies. The specific equations of mass, momentum and energy conservation solved by FLAG for the bulk materials can be found in [41] and can be written for two-way coupled, particle-laden flow as

$$\rho \frac{d\rho^{-1}}{dt} - \nabla \cdot \mathbf{v} = 0 \tag{58}$$

$$\rho \frac{d\mathbf{v}}{dt} + \nabla P = -\frac{1}{V_z} \sum_{i=1}^{N_p} \xi_i \mathbf{F}_i^{pg} \tag{59}$$

$$\rho \frac{de}{dt} + P\nabla \cdot \mathbf{v} = -\frac{1}{V_z} \sum_{i=1}^{N_p} \xi_i (G_i^{pg} + Q_i^{pg}) \,. \tag{60}$$

In the above, ρ is the material density, $\mathbf{v}$ is the mesh velocity, P is pressure and e is the specific internal energy. FLAG is capable of using arbitrary Lagrangian-Eulerian (ALE) mesh relaxers if necessary to avoid mesh tangling. It should be noted that Eqn. 58 is not explicitly solved by the code as it is immediately satisfied by definition in a Lagrangian framework. The right-hand side terms in Eqns. 59 and 60 arise from the two-way coupling of hydrodynamic drag and heat transfer from ejecta particles in a given zone z of volume V_z to the carrier fluid where $\mathbf{F}^{pg}$ is the drag, $G^{pg} = \mathbf{F}^{pg} \cdot \mathbf{v}$ is the work due to drag and Q^{pg} is the heat transfer between the phases. Lastly, ξ is the number of physical particles represented by a computational point-particle. The equations of state used for thermodynamic closure are either the ideal gas law for the fluid species, with the vacuum cases handled as a gas with extremely low density and pressure ($O(100Pa)$), or the SESAME 90601 table for cerium.

In FLAG, ejecta particles, once produced, are treated as Lagrangian point-particles. Thus, each computational ejecta particle has its motion and temperature tracked individually. In this context, a computational particle represents a group of physical particles which are assumed to have the same physical properties as well as position and velocity vectors. This concept is utilized for both feasibility concerns as well as for reduction in computational costs since simulating every physical particle in an ejecta cloud is usually intractable. The equations governing the ejecta state for a computational particle i is given as

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{u}_i \tag{61}$$

$$m_{p,i} \frac{d\mathbf{u}_i}{dt} = \mathbf{F}_i^{pg} \tag{62}$$

$$m_{p,i} c^m \frac{dT_{p,i}}{dt} = Q_i^{pg} \,. \tag{63}$$

In these equations, $\mathbf{x}$ and $\mathbf{u}$ are the particle position and velocity vectors. Additionally, m_p, c^m and T_p are the particle's mass, specific heat and temperature respectively. Comparing Eqns. 30

and 63 gives us that Q^{pg} is nothing more than the right hand side of the model equation given in Eqn. 30 (in other words, the convective and reactive heat transfer terms). The drag model used to obtain $\mathbf{F}^{pg}$ is given by Parmar *et al.* [47] and is applicable to the compressible flow regimes to be seen in the simulations. The above two sets of equations are solved alongside Eqs. 27 (or 21) and 30 to complete the equation set.

A final note on the ejecta modeling in FLAG is the model used to create the ejecta particles from the metal target after shock release. The sourcing model used is described in detail in [13]. Briefly, the source model assumes that the target is melted upon shock release and undergoes RM instability. The code solves a differential equation to "resolve" the instability growth and birth ejecta particles once the tips of the growing spikes are assumed to hydrodynamically break off from the spike as droplets. The new particles, by default, are assumed to take the spike tip velocity as their velocity and inherit the thermodynamic properties of the material zones behind the faces they are sourced from. Additional velocity distributions which are empirical and based off experimental data can be used on top of this assumption. In this work, the undropped self-similar velocity distribution (SSVD) was used for the solid ejecta calibration simulations while the dropped SSVD model was used for all other simulations.

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